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EXPERIMENT NO 5

Aim: To automate application deployment using Github/ Jenkins.

Theory:

Steps:

1. Create a GitHub account.
2. Create a GitHub repository.
3. Create a sample application.
4. Push the application to GitHub.
5. Install Java JDK.
6. Install Jenkins.
7. Start Jenkins on the local machine.
8. Create a Jenkins administrator account.
9. Create a Jenkins Pipeline.
10. Connect Jenkins with the GitHub repository.
11. Create a Jenkinsfile.
12. Configure Build, Test and Deploy stages.
13. Execute the Jenkins pipeline.
14. Verify the deployment.
15. Configure a GitHub webhook for automatic triggering.

- Check Java is installed on your machine
- Run java –version

- Open the official Jenkins Windows installation page:
  Jenkins Windows Installation Guide
- Download the **Jenkins LTS Windows installer (.msi)**.
- Run the installer **as Administrator**:
- Right-click the downloaded .msi
- Select **Run as administrator**
- Complete the installation

  Press windows + R

  Services.msc

  See-> Jenkins with status Running

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Then open

https://localhost:8080

If Jenkins id running then create your first pipeline

Step 1: Open Jenkins

Open your browser and go to:

http://localhost:8080

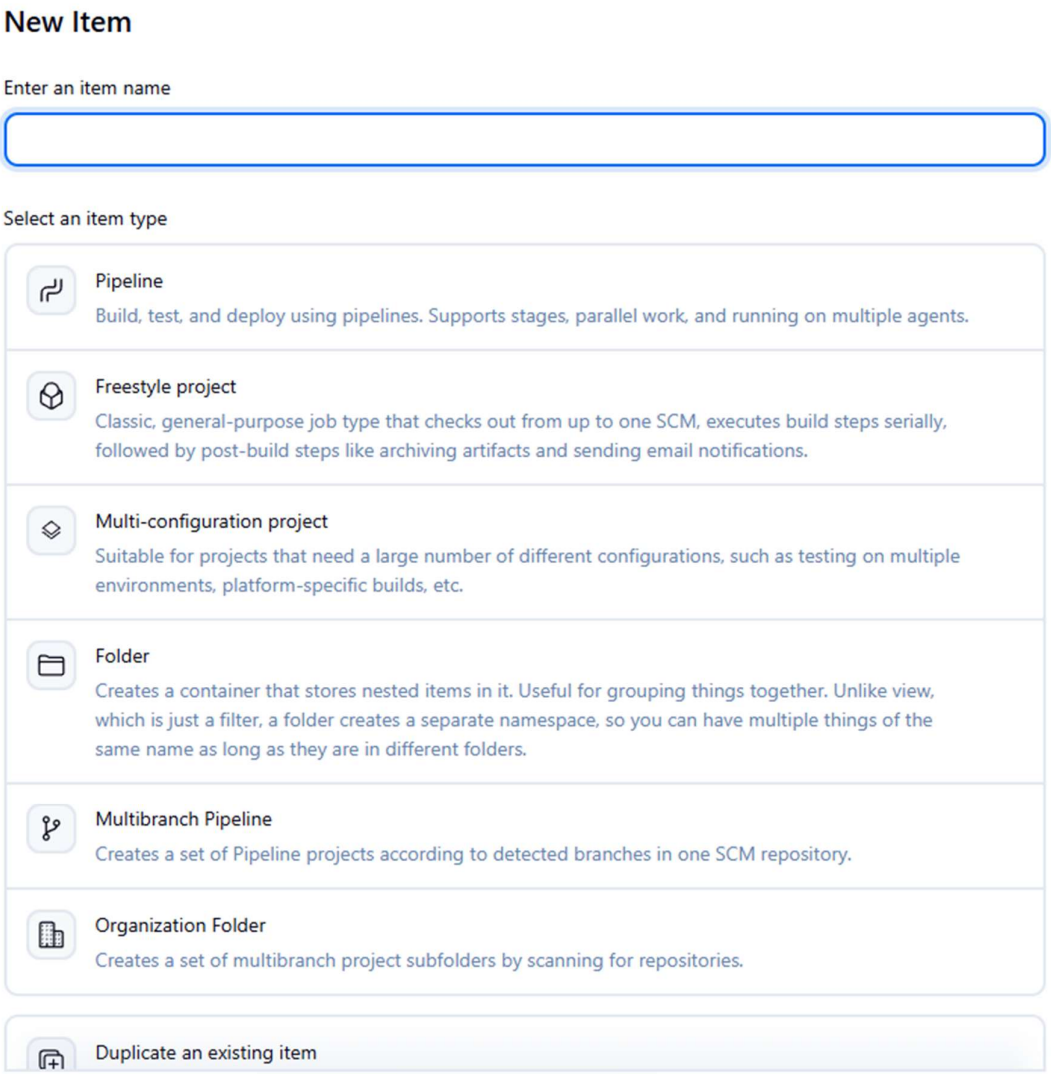
You should see the **Jenkins Dashboard**.

Step 2: Click "New Item"

On the left side of the Jenkins Dashboard, click:

New Item

You will get a screen similar to



Enter any name for e.g. GitHub-Jenkins-Demo

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Then select:

Pipeline

Click: **OK**

Step 3: Find the Pipeline section

A configuration page will open.

Don't change anything else for now.

Scroll toward the bottom until you find:

Pipeline

You will see a large text box called:

Script

We are going to put a very small Pipeline in it.

Step 4: Enter this Pipeline

Copy this exactly:

```
pipeline {
    agent any

    stages {

        stage('Build') {
            steps {
                echo 'Building the application...'
            }
        }

        stage('Test') {
            steps {
                echo 'Testing the application...'
            }
        }
```

```
        stage('Deploy') {
            steps {
                echo 'Deploying the application...'
            }
        }
    }
}
```

This is a **Declarative Jenkins Pipeline**. Jenkins defines pipeline, agent, stages, and steps as core elements of this syntax.

Step 5: Save

Scroll to the bottom.

Click:

Save

You should now be taken to the page for:

GitHub-Jenkins-Demo

Part 2 — Run the Pipeline

On the left side, click:

Build Now

You should see a build appear under:

Build History

Something like:

#1

Click:

#1

Then click:

Console Output

You should see messages similar to:

Building the application...

Testing the application...

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Deploying the application...

Finished: SUCCESS

This means your Jenkins Pipeline is working.

We have first verified Jenkins can execute pipeline on the laptop. Once this works, we'll connect GitHub.

Part 4 — What each stage means

Build

stage('Build')

Means:

Prepare/create the application.

For a real Java application, this could compile the Java source code.

For a Python application, it could install dependencies or package the application.

For our simple HTML application, there is essentially nothing to compile.

Test

stage('Test')

Means:

Check whether the application works correctly.

Later we can put actual automated tests here.

Deploy

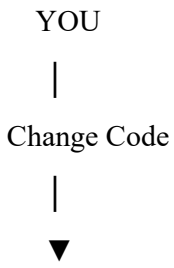
stage('Deploy')

Means:

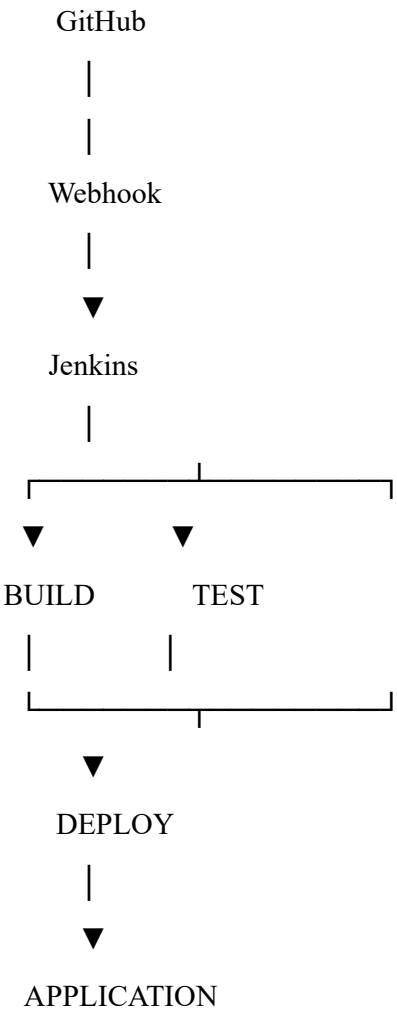
Put the successfully tested application into its target environment.

For our beginner experiment, we'll eventually copy the HTML application to a deployment folder on your laptop.

Final Goal is

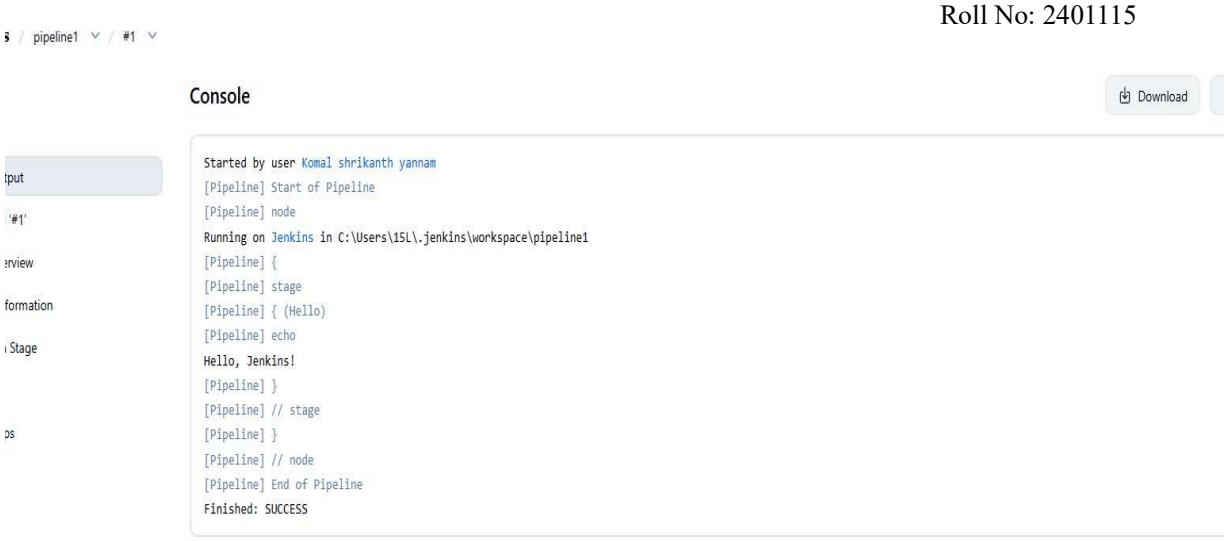


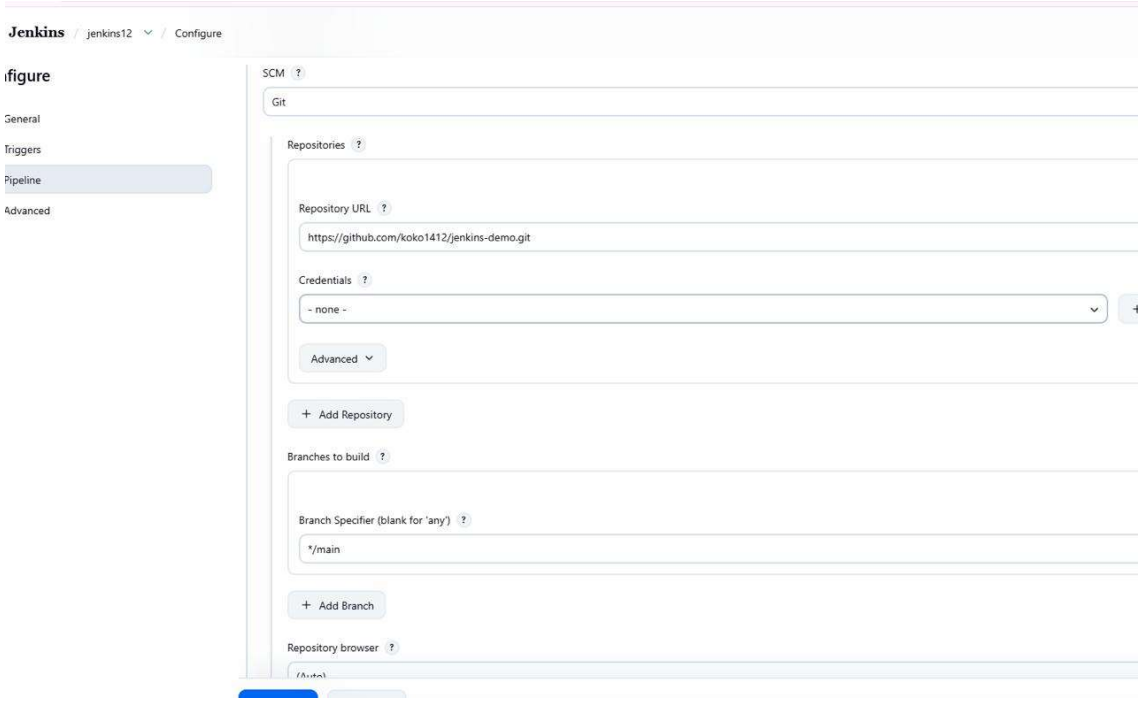
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OUTPUT:

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